Hyeon Jeon

Ph.D. Student, Department of Computer Science and Engineering, Seoul National University
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Education

2022 – Present **Ph.D. Student in Computer Science and Engineering**, Seoul National University
Advised by Dr. Jinwook Seo
including alternative military service as a technical research personnel (2024 – Present)

2020 – 2022 **Master's Student in Computer Science and Engineering**, Seoul National University
Switched to Ph.D. program in Sep. 2022

2016 – 2020 **Bachelor of Science in Computer Science and Engineering**, POSTECH
Early Graduation with Honors (seven semesters, Magna Cum Laude)

Research Interests

Hyeon's research lies at the intersection of data visualization and machine learning. His primary goal is to make visual analytics more reliable. To this end, he develops machine learning algorithms, quality metrics, and interaction techniques for visual analytics.

Publications

Peer-Reviewed Conference Full Papers & Journal Articles

The papers in the IEEE VIS proceedings are published as a special issue of TVCG.
Papers published in TVCG are invited for presentation at related conferences like VIS.
* denotes equal contribution.

2024 P9 **Phenoflow: A human-llm driven visual analytics system for exploring large and complex stroke datasets**
Jaeyoung Kim, Sihyeon Lee, **Hyeon Jeon**, Keon-Joo Lee, Hee-Joon Bae, Bohyoung Kim, and Jinwook Seo
IEEE Transactions on Visualization and Computer Graphics (TVCG, Proc. VIS 2024)

P8 **Natural language dataset generation framework for visualizations powered by large language models**
Hyung-Kwon Ko, **Hyeon Jeon**, Gwanmo Park, Dae Hyun Kim, Nam Wook Kim, Juho Kim, and Jinwook Seo
ACM Conference on Human Factors in Computing Systems (CHI 2024)

P7 **CloChat: Understanding how people customize, interact, and experience personas in large language models**
Juhye Ha, **Hyeon Jeon**, DaEun Han, Jinwook Seo, and Changhoon Oh
ACM Conference on Human Factors in Computing Systems (CHI 2024)

2023 P6 **MoNetExplorer: A visual analytics system for analyzing dynamic networks with temporal network motifs**
Seokweon Jung, DongHwa Shin, **Hyeon Jeon**, Kiroong Choe, and Jinwook Seo
IEEE Transactions on Visualization and Computer Graphics (TVCG)
Invited to IEEE VIS 2024 for presentation

P5 **Classes are not clusters: Improving label-based evaluation of dimensionality reduction**
Hyeon Jeon, Yun-Hsin Kuo, Michael Aupetit, Kwan-Liu Ma, and Jinwook Seo
IEEE Transactions on Visualization and Computer Graphics (TVCG, Proc. VIS 2023)

P4 **CLAMS: A cluster ambiguity measure for estimating perceptual variability in visual clustering**
Hyeon Jeon*, Ghulam Jilani Quadri*, Hyunwook Lee, Paul Rosen, Danielle Albers Szafir, and Jinwook Seo
Transactions on Visualization and Computer Graphics (TVCG, Proc. VIS 2023)
Best Papers Honorable Mention Awards (top 5% submissions)

P3 **Large-scale text-to-image generation models for visual artists' creative works**
ACM International Conference on Intelligent User Interfaces (IUI 2023)
Hyung-Kwon Ko, Gwanmo Park, **Hyeon Jeon**, Jaemin Jo, Juho Kim, and Jinwook Seo

2021 P2 **Measuring and explaining the inter-cluster reliability of multidimensional projections**
IEEE Transactions on Visualization and Computer Graphics (TVCG, Proc. VIS 2021)
Hyeon Jeon, Hyung-Kwon Ko, Jaemin Jo, Youngtaek Kim, and Jinwook Seo

2020 P1 **Githru: Visual analytics for understanding software development history through git metadata analysis**
IEEE Transactions on Visualization and Computer Graphics (TVCG, Proc. VAST 2020)
Youngtaek Kim, Jaeyoung Kim, **Hyeon Jeon**, Young-Ho Kim, Hyunjoo Song, Bohyoung Kim, and Jinwook Seo

Peer-Reviewed Conference Short Papers

2024 S6 **CLeVer: Continual learning visualizer for detecting task transition failure**
Minsuk Chang, Donghun Kim, **Hyeon Jeon**, Seokweon Jung, and Jinwook Seo
IEEE Pacific Visualization Conference (PacificVis 2024)

S5 **IoLens: Visual analytics system for exploring storage i/o tracking process**
Changmin Jeon, Jiwon Ha, Hyolim Hong, **Hyeon Jeon**, Hyeonsang Eom, Heonyoung Yeom, and Jinwook Seo
IEEE Pacific Visualization Conference (PacificVis 2024)

2023 S4 **ZADU: A python library for evaluating the reliability of dimensionality reduction embeddings**
Hyeon Jeon, Aeri Cho, Jinhwa Jang, Soohyun Lee, Jake Hyun, Hyung-Kwon Ko, Jaemin Jo, and Jinwook Seo
IEEE Visualization and Visual Analytics (VIS 2023)

2022 S3 **Uniform manifold approximation with two-phase optimization**
Hyeon Jeon*, Hyung-Kwon Ko*, Soohyun Lee, Jaemin Jo, and Jinwook Seo
IEEE Visualization and Visual Analytics (VIS 2022)

S2 **VANT: A visual analytics system for refining parallel corpora in neural machine translation**
Sebeom Park, Soohyun Lee, Youngtaek Kim, **Hyeon Jeon**, Seokweon Jung, Jinwook Bok, and Jinwook Seo
IEEE Pacific Visualization Symposium (PacificVis 2022)
Best Visualization Notes (top 1st submission)

2021 S1 **Visualization support for multi-criteria decision making in software issue propagation**
Youngtaek Kim, **Hyeon Jeon**, Young-Ho Kim, Yuhoon Ki, Hyunjoo Song, and Jinwook Seo
IEEE Pacific Visualization Symposium (PacificVis 2021)

Panel Organizations

2024 A1 **How many evaluations are enough? A panel discussion on evaluation trends in information visualization**
Ghulam Jilani Quadri, Danielle Albers Szafir, Arran Zeyu Wang, and **Hyeon Jeon**
IEEE Visualization and Visual Analytics (VIS 2024)

Workshop Papers, Posters, and Domestics

2024 O5 **Efficiently crowdsourcing visual importance with punch-hole annotation**
Minsuk Chang, Soohyun Lee, Aeri Cho, **Hyeon Jeon**, Seokhyeon Park, Cindy Xiong Bearfield, and Jinwook Seo
IEEE Visualization and Visual Analytics (VIS 2024), Poster

O4 **Combinational nonuniform timeslicing of dynamic networks**
Seokweon Jung, DongHwa Shin, **Hyeon Jeon**, and Jinwook Seo
IEEE Pacific Visualization Conference (PacificVis 2024), Poster

2023 O3 **A vega-lite dataset and natural language generation pipeline with large language models**
Hyung-Kwon Ko, **Hyeon Jeon**, Gwanmo Park, Dae Hyun Kim, Nam Wook Kim, Juho Kim, and Jinwook Seo
NLVIZ 2023 Workshop: Exploring Research Opportunities for NL, Text, and Data Visualization

O2 **Interactive visual analytics system for criminal intelligence analysts with multiple coordinated views**
Seokweon Jung, DongHwa Shin, Jinwook Bok, Seokhyeon Park, **Hyeon Jeon**, Jinwook Seo, Insoo Lee, and Sooyoung Park
Journal of KIISE

2021 O1 **Interactive visualization for exploring information fragments in software repositories**
IEEE Pacific Visualization Symposium (PacificVis 2021), Poster

Awards and Honors

Paper Awards

2024 **Best Papers Honorable Mentions**, IEEE VIS 2023
Top 5% submissions among all full paper submissions

2022 **Best Visualization Notes**, IEEE PacificVis 2022
Top 1st submission among all short paper submissions

Fellowships and Scholarships

2024 **Future Innovation Individual (Gold Award)**, SNU BK21 Four
1st place among student researchers participating in SNU BK21 Four
Granted 3,000,000 KRW (≃ 2,700 USD)

2023 **Outstanding Research Individual Fellowship**, SNU BK21 Four, Dept. of CSE
Granted 10,000,000 KRW (≃ 9,000 USD)

2023 **Start Student Researcher Award**, SNU BK21 Four, Dept. of CSE
Granted 700,000 KRW (≃ 600 USD)

2016 – 2019 **Jigok Scholarship**, POSTECH
Full scholarship throughout undergraduate (Seven semesters)

Travel Grants

2024 **Conference Travel Award**, Korea Computer Congress
Support for conference registration and travel expenses

Acknowledgements for Community Services

2024 **Special Recognitions for Outstanding Reviews**, PacificVis 2024

Academic Services

Program Committee

2024 Video Showcase, ACM CHI Conference on Human Factors in Computing Systems (CHI)

Conference Organizing Committee

2024 **Student Volunteer Chair**, IEEE Visualization and Visual Analytics (VIS)

Student Volunteers

2022 – 2024 IEEE Visualization and Visual Analytics (VIS)
2023 IEEE Pacific Visualization Symposium (PacificVis)
2022 ACM CHI Conference on Human Factors in Computing Systems (CHI)

Conference Reviewer (External)

2022 – 2024 IEEE Visualization and Visual Analytics (VIS)
2022 – 2024 Eurographics/IEEE VGTC Conference on Visualization (EuroVis)
2022 – 2024 IEEE Pacific Visualization Symposium (PacificVis)
2022 – 2024 ACM SIGCHI Conference on Human Factors in Computing Systems (CHI)
2023 ACM Conference on Intelligent User Interfaces (IUI)
2024 ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW)
2022 Conference on Neural Information Processing System (NeurIPS)

Invited Journal Reviewer

2023 – 2024 Journal of Visualization and Interaction (JoVI)
2024 IEEE Transactions on Geoscience and Remote Sensing (TGRS)
2023 ACM Transactions on Knowledge Discovery from Data (TKDD)
2021 – 2022 Visual Informatics

Funded Grants

2024 – 2026 **GuardVis: AI-Powered Validation for Secure and Reliable Data Analysis and Visualization**,
National Research Foundation of Korea, Ph.D. Research Fellowship, No. RS-2024-00409716
Investigator: **Hyeon Jeon**
Amount: 40,000,000 KRW (≃ 30,000 USD)
Accepted, but withdrawn due to my alternative military service as a technical research personnel

Experiences

Research Experiences

2022 **Visiting Ph.D. Student**, University of California, Davis (advised by Dr. Kwan-Liu Ma)
2020 **Undergraduate Intern**, Human-Computer Interaction Lab., Seoul National University
2019 **Undergraduate Intern**, Algorithms Lab, POSTECH
2018 **Undergraduate Intern**, Computer Vision Lab, POSTECH

Teaching Experiences

Teaching Assistants

2021 – 2024 **Information Visualization and Visual Analytics**, Seoul National University
Contributed to course materials and programming assignments (D3.js)
Semi-advised three team projects to be conference papers [S2, S5, S6]

2021 **Digital Computer Concept and Practice**, Seoul National University
Contributed to course materials and programming assignments (Python)

2023 **Samsung AI Expert Program**, Seoul National University
Course title: Visual analytics for high-dimensional data
Designed the course syllabus and programming practice session (Python)
ZADU [S4] was used for the programming practice

2020 – 2021 **Samsung DS^2 Program**, Seoul National University
Contributed to course materials and programming assignments (Altair)

Tutoring and Mentoring

2021, 2023 **Undergraduate Research Opportunities Program (UROP)**, Seoul National University
2019, 2020 **CSE Student Mentoring Program (SMP)**, POSTECH

Research Mentoring with Artifacts

All individuals listed have consented to be named in my CV.

2023 **Minsuk Chang**, CSE Undergraduate Student, Seoul National University
Co-authored [S6], [05]. Currently a Ph.D. student at Georgia Institute of Technology

2023 – Present **Jinhwa Jang**, CSE Master's Student, Seoul National University
Co-authored [S4]. Currently a master's student at Seoul National University

2021 – Present **Soohyun Lee**, CSE MS/Ph.D. Student, Seoul National University
Co-authored [S4], [S3], [S2]. Currently a Ph.D. student at Seoul National University

2022 **Sebeom Park**, CSE Master's Student, Seoul National University
Co-authored [S2]. Currently a staff engineer at Samsung Electronics

Invited Talks

2024 **Towards more reliable high-dimensional data visualization**
POSTECH (HIS Lab), July 2024
University of California, Davis (Visualization Research Group), May 2024
Georgia Institute of Technology (Visualization Group), April 2024
KAIST (Kixlab), February 2024
Google, February 2024
Carnegie Mellon University (Data Interaction Group), February 2024

2022 – 2023 **Making clusters more reliable**
Sungkyunkwan University (Interactive Data Computing Lab), August 2023
UNIST (HAiV Lab Visualization Seminar 2023), August 2023
Sejong University (Data Visualization Lab), May 2023
KAIST (Kixlab), February 2024
Google, February 2024
University of North Carolina, Chapel Hill (Dept. of Computer Science) November 2022

2024 **Ambiguity in data visualizations**

UNIST (HAiV Lab Visualization Seminar 2024) July 2024
UNIST (Guest Lecture, Introduction to Human-Computer Interaction [CSE333]), May 2024

2024 **Classes are not clusters: Improving label-based evaluation of dimensionality reduction**
Korea Computer Congress, June 2024

2023 **CLAMS: A cluster ambiguity measure for estimating perceptual variability in visual clustering**
Korea Software Congress, December 2023
Seoul National University (Computing Frontier Summer School), August 2023

2022 **Measuring and explaining the inter-cluster reliability of multidimensional projections**
Korea Software Congress, December 2022

Press Coverage

2023 **NAVER-SNU joint research, honored as 'Top 5%' at the prestigious data analysis conference**
The Electronic Times (Korean)

2022 **Measuring and explaining the inter-cluster reliability of multidimensional projections**
Seoul National University Artificial Intelligence Institute (SNU AIIS) Tech Blog

References

Dr. Jinwook Seo, Professor, Seoul National University
jseo@snu.ac.kr

Dr. Kwan-Liu Ma, Distinguished Professor, University of California, Davis
klma@ucdavis.edu

Dr. Michaël Aupetit, Senior Scientist, QCRI, Hamad Bin Khalifa Univiversity
maupetit@hbku.edu.qa

Dr. Ghulam Jilani Quadri, Assistant Professor, University of Oklahoma
quadri@ou.edu

Dr. Jaemin Jo, Assistant Professor, Syunkyunkwan University
jmjo@g.skku.edu